

Md Hassan

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

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Profile

M.Tech graduate (IIT Kanpur) specializing in AI-driven cybersecurity, encrypted traffic analysis, and real-time machine learning systems. Designed end-to-end ML pipelines achieving 90%+ accuracy on VPN detection with a 25% reduction in false positives. GATE-qualified in Computer Science and Data Science & AI (consecutive years); UGC-NET qualified. Seeking a research or engineering role at the intersection of Machine Learning, Network Security, and AI Systems.

Education

Indian Institute of Technology (IIT) Kanpur

June 2026

- Master of Technology (M.Tech.) — PSE · Kanpur, India
- Focus: Machine Learning Systems, Applied AI, Intelligent Infrastructure

Maulana Azad National Urdu University (MANUU)

May 2026

- B.Tech. in Computer Science · **CGPA: 8.96 / 10** · Hyderabad, India
- Coursework: Machine Learning, Deep Learning, Network Security, Data Structures & Algorithms, Computer Networks, Cryptography, Operating Systems, Database Systems

Research & Technical Projects

Multi-Layered VPN Detection & Traffic Deanonymisation

2025 – Present

LEAD RESEARCHER

- Engineered a hybrid **Deep Packet Inspection (DPI) + Machine Learning** pipeline classifying encrypted VPN tunnels across 5+ protocol types (OpenVPN, WireGuard, Shadowsocks, I2P).
- Achieved **90%+** protocol-level classification accuracy using flow metadata fusion, statistical fingerprinting, and ensemble feature selection.
- Reduced false positive rate by **25%** through multi-stage filtering, behavioral flow analysis, and per-protocol threshold calibration.
- Built high-throughput packet parsing pipeline with Scapy, processing 10,000+ packet flows per session for real-time analysis.

AI-Powered Intrusion Detection System (IDS)

2024

ML ENGINEER

- Designed and deployed a **real-time ML-based IDS** for high-throughput network threat classification and anomaly detection using Python and Scikit-learn.
- Applied feature engineering on NetFlow/PCAP data to improve detection precision; optimized inference pipeline to **sub-10 ms latency** using containerized Docker deployment.
- Benchmarked against signature-based IDS baselines, demonstrating measurable ML uplift across attack categories.

M-ACT: Mirrored Active Collaborative Tunnel

2025

SYSTEMS DEVELOPER

Learn2Play: AI-Driven Adaptive Learning System

2024 – 2025

AI DEVELOPER

- Built an **ML-based adaptive content recommendation engine** using behavior modeling, real-time engagement signals, and data-driven personalization.
- Applied reinforcement learning principles to optimize long-term content sequencing; integrated behavioral analytics for continuous model refinement.

Industry Experience

FlyRank AI

July 2026 – Present

ML ENGINEERING INTERN

Remote

- Building and optimizing production **ML models and scalable inference workflows** for AI-driven marketing analytics and content optimization.
- Contributing to end-to-end ML pipelines: data ingestion, feature engineering, model training, serving, and performance monitoring.

India Space Lab

May 2026 – June 2026

SPACE TECHNOLOGY INTERN

Remote

- Completed intensive technical training in CanSat/CubeSat systems, Remote Sensing, GIS analysis, Rocketry Science, and Disaster Management protocols.
- Applied systems engineering and data analysis methods to satellite mission design workflows.

Skillship Hyderabad

Jan 2023 – Jan 2024

DESIGN ENGINEER INTERN

Hyderabad, India

- Delivered cybersecurity awareness and networking fundamentals training to **100+ students**, covering TCP/IP, protocol dissection, and vulnerability assessment.
- Designed hands-on lab exercises for network security analysis and penetration testing; developed learning modules with measurable competency outcomes.

Skills

AI / ML	Scikit-learn, Feature Engineering, Anomaly Detection, Classification, Real-time Inference, Adaptive Systems, Reinforcement Learning
Cybersecurity	Deep Packet Inspection (DPI), VPN Fingerprinting, IDS/IPS, Intrusion Detection, Network Forensics, Traffic Analysis, Email Forensics, SIEM
Programming	Python (Expert), C/C++, Bash/Shell Scripting
Frameworks	FastAPI, Scapy, Docker, Jupyter, Git, Wireshark, Nmap, Metasploit, Splunk
Networking	TCP/IP, UDP, NetFlow, PCAP, OpenVPN, WireGuard, HTTP/S, DNS, TLS/SSL

Scholarships & Achievements

- **GATE 2026** — Qualified in Computer Science (CS) and Data Science & AI (DA), IIT Guwahati

- **GATE 2025** — Qualified in Computer Science (CS), IIT Roorkee
- **UGC NET 2025** — Qualified in Computer Science & Applications
- **CGPA 8.96 / 10** — Top standing, B.Tech. Computer Science, MANUU Hyderabad

Leadership & Service

Entrepreneurship Cell (E-Cell), MANUU Hyderabad

Aug 2023 – May 2026

FOUNDING MEMBER

Hyderabad, India

- Co-founded the university's inaugural E-Cell, building a student innovation ecosystem supporting AI, cybersecurity, and data-driven startup initiatives.
- Mentored 5+ student teams on technology project development and startup ideation; organized interdisciplinary innovation workshops.